

c) Test $H_0: p = 0.3$ vs. $H_a: p < 0.3$

04/27
 $\alpha = 0.05$

M. ①

① test statistic

$$Z^* = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.22 - 0.3}{\sqrt{\frac{0.3(1-0.3)}{100}}} = -1.75$$

② critical value.

$$-Z_{\alpha} = -Z_{0.05} = -1.645$$

③ Decision and conclusion.

$$Z^* = -1.75 < -1.645 = -Z_{0.05}, \quad \underline{\text{reject } H_0}$$

We have strong enough evidence to suggest $p < 0.3$.

d) p-value for test in c)

$$p\text{-value} = P(Z < -1.75) = 0.0401$$

$$p\text{-value} = 0.0401 < 0.05 = \alpha, \quad \underline{\text{reject } H_0}$$

Chi-squared test for independence

(2)

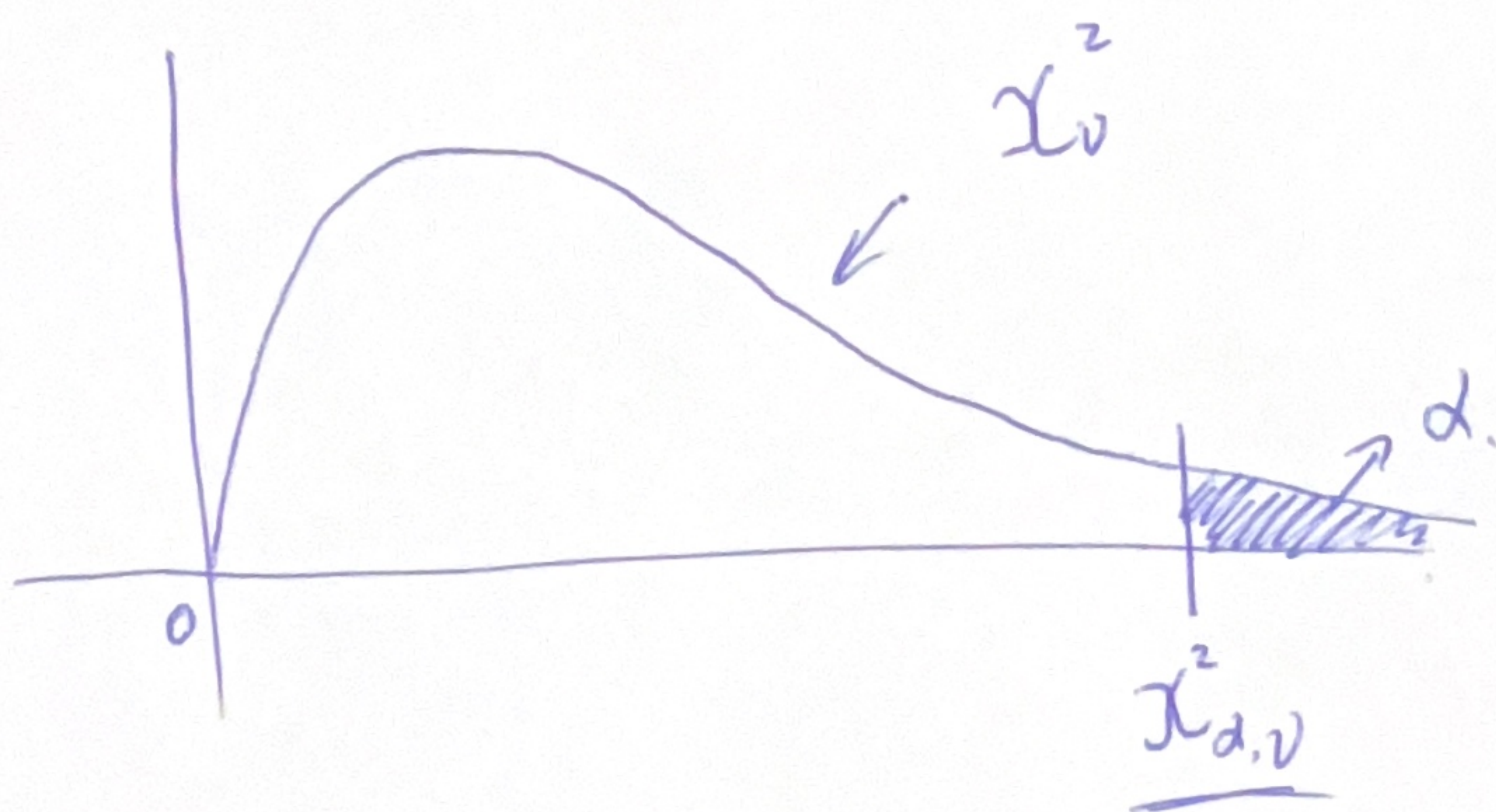
2. Categorical variables, then single sample from single population.
Then test if two variables independent or not.

Def'n: A chi-squared distribution has support $(0, \infty)$

$\chi^2_{\underset{\substack{\uparrow \\ \text{df}}}{v}}$ - chi-squared distribution with df or v .

Def'n: Let $\chi^2_{\alpha, v}$ be the point on the number line such that.

$$P(\chi^2_v > \chi^2_{\alpha, v}) = \alpha.$$



Test: H_0 : A and B are independent.

vs. H_a : A and B are not independent.

Example: 793 bicycle accidents.

	Helmet	No Helmet	row total
Health insurance	17 (43.6)	218 (191.4)	235
No health insurance	130 (103.4)	428 (454.6)	558
column total	147	646	793 grand total

a) Test H_0 : Hel and HI are independent (not associated) (3)

vs. H_a : ... are not independent (associated)

$$\alpha = 0.05.$$

b) Expected counts.

$$\begin{aligned} e_{ij} &= \text{expected count in row } i \text{ and column } j. \\ &= \frac{\text{row total} \times \text{column total}}{\text{grand total}}. \end{aligned}$$

$$e_{11} = \frac{235 \times 147}{793} = 43.6$$

$$e_{12} = \frac{235 \times 646}{793} = 191.4$$

$$e_{21} = \frac{558 \times 147}{793} = 103.4$$

$$e_{22} = \frac{558 \times 646}{793} = 454.6$$

1) test statistic.

$$\begin{aligned} \chi^2 &= \sum \frac{(o_{ij} - e_{ij})^2}{e_{ij}} = \sum \frac{(o - e)^2}{e} \\ &= \frac{(17 - 43.6)^2}{43.6} + \frac{(218 - 191.4)^2}{191.4} + \frac{(130 - 103.4)^2}{103.4} \\ &\quad + \frac{(428 - 454.6)^2}{454.6} = 28.255 \end{aligned}$$